

Single cell methods in epigenomics

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Workshop – Epigenomics Data Analysis

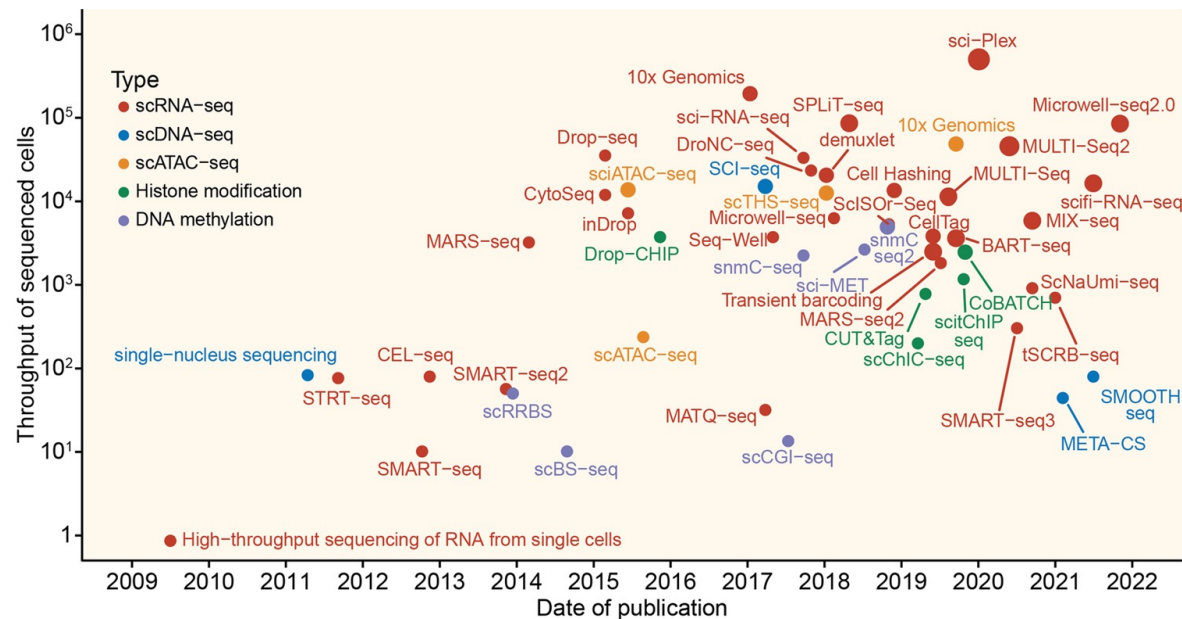
Single cell methods

- Analyzing bulk samples will give results that represent an average of the cell population, over thousands or millions of cells.
- Single cell methods give results for each cell.
 - Differences between individual cells (due to stochasticity, cell cycle etc).
 - Differences between cell types.
- More noise in data.
- Require new analysis steps.

This is a big and rapidly changing field. We have a whole course on single cell RNA-seq. In this talk we will give a short introduction to single cell methods in epigenomics.

Today's talk

- Background on single cell methods
 - Single cell ATAC-seq
 - Single cell data analysis, common steps
 - Single cell ChIP-seq & CUT&TAG
 - Single cell DNA-methylation
 - Spatial methods
-
- Broad overview
 - Focus on concepts over details.



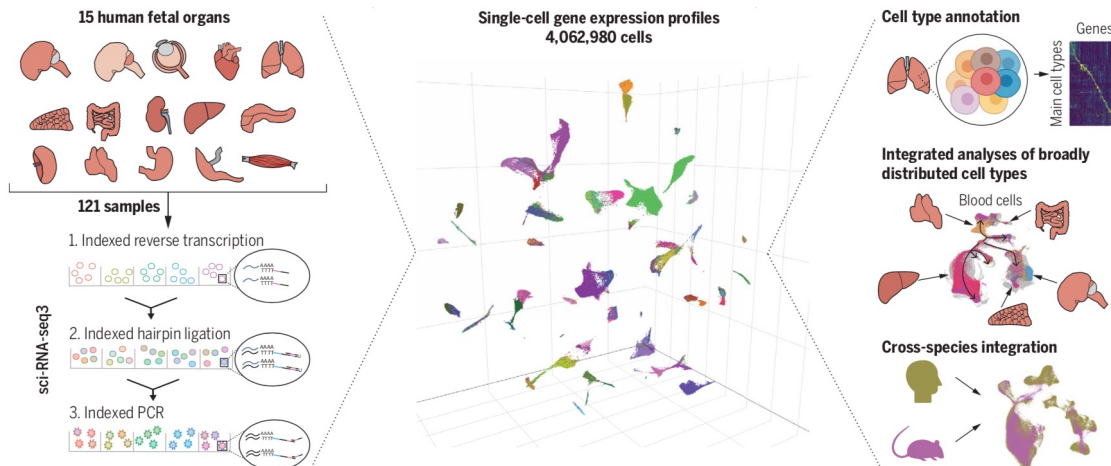
Example study (2020)

RESEARCH ARTICLE

HUMAN GENOMICS

A human cell atlas of fetal gene expression

Junyue Cao^{1*}, Diana R. O'Day², Hannah A. Pliner³, Paul D. Kingsley⁴, Mei Deng², Riza M. Daza¹, Michael A. Zager^{3,5}, Kimberly A. Aldinger^{2,6}, Ronnie Blecher-Gonen¹, Fan Zhang⁷, Malte Spielmann^{8,9}, James Palis⁴, Dan Doherty^{2,3,6}, Frank J. Steemers⁷, Ian A. Glass^{2,3,6}, Cole Trapnell^{1,3,10}†, Jay Shendure^{1,3,10,11}†



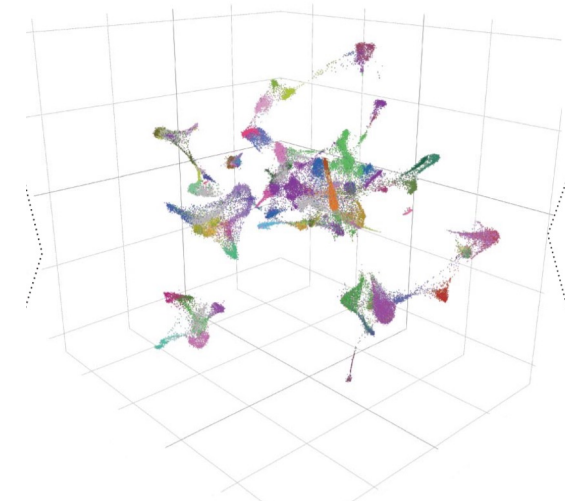
RESEARCH ARTICLE SUMMARY

HUMAN GENOMICS

A human cell atlas of fetal chromatin accessibility

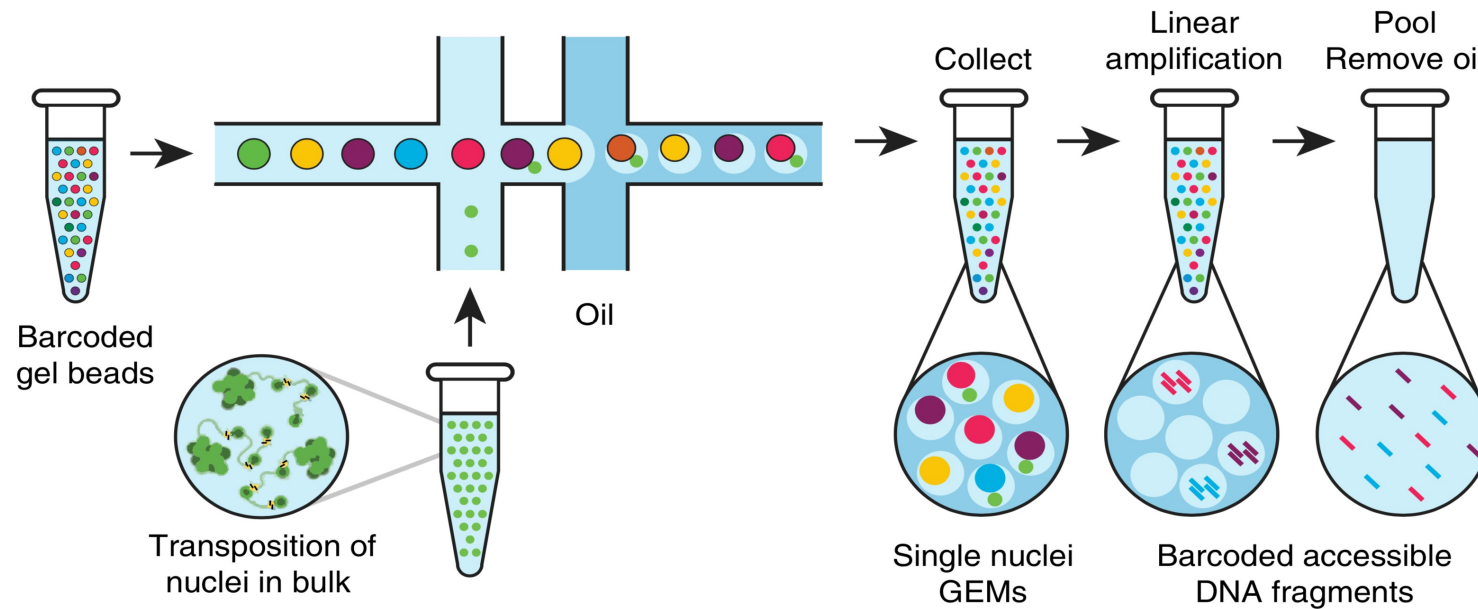
Silvia Domcke*, Andrew J. Hill*, Riza M. Daza*, Junyue Cao, Diana R. O'Day, Hannah A. Pliner, Kimberly A. Aldinger, Dmitry Pokholok, Fan Zhang, Jennifer H. Milbank, Michael A. Zager, Ian A. Glass, Frank J. Steemers, Dan Doherty, Cole Trapnell†, Darren A. Cusanovich†, Jay Shendure†

Single-cell chromatin accessibility profiles
790,957 cells



Single cell ATAC-seq

- First paper, from Greenleaf lab: (Buenrostro et al. 2015, Nature)
- Now available as a kit from 10X Genomics
 - Each nuclei is attached to a bead containing a different barcode, inside an oil droplet.
 - These barcodes are attached to the DNA fragments, making it possible to assign each sequenced DNA fragment to a cell.

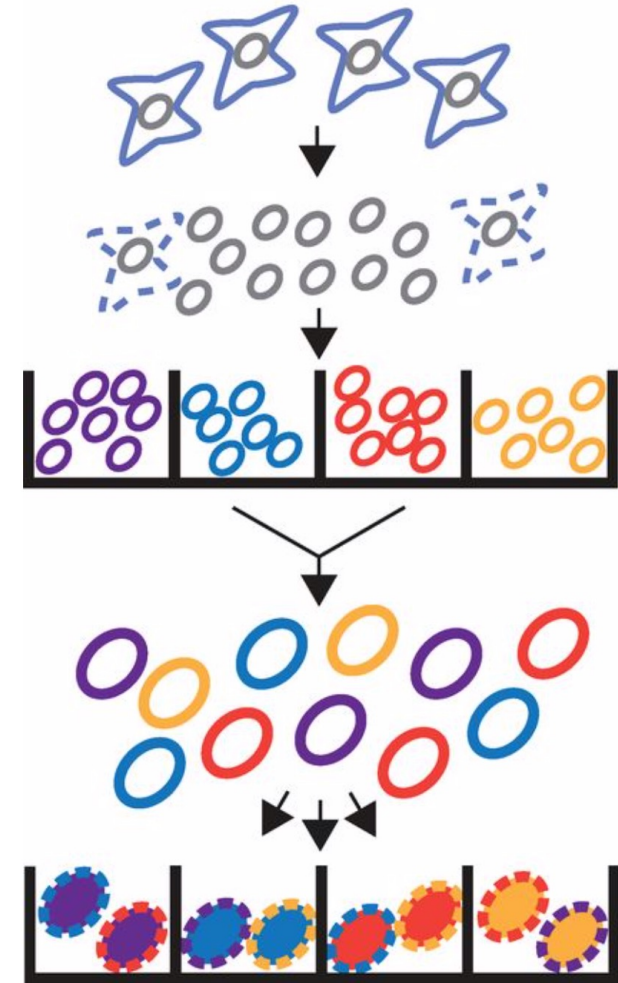


(Satpathy et al. 2019 Nature Biotechnology)



Single cell ATAC-seq II

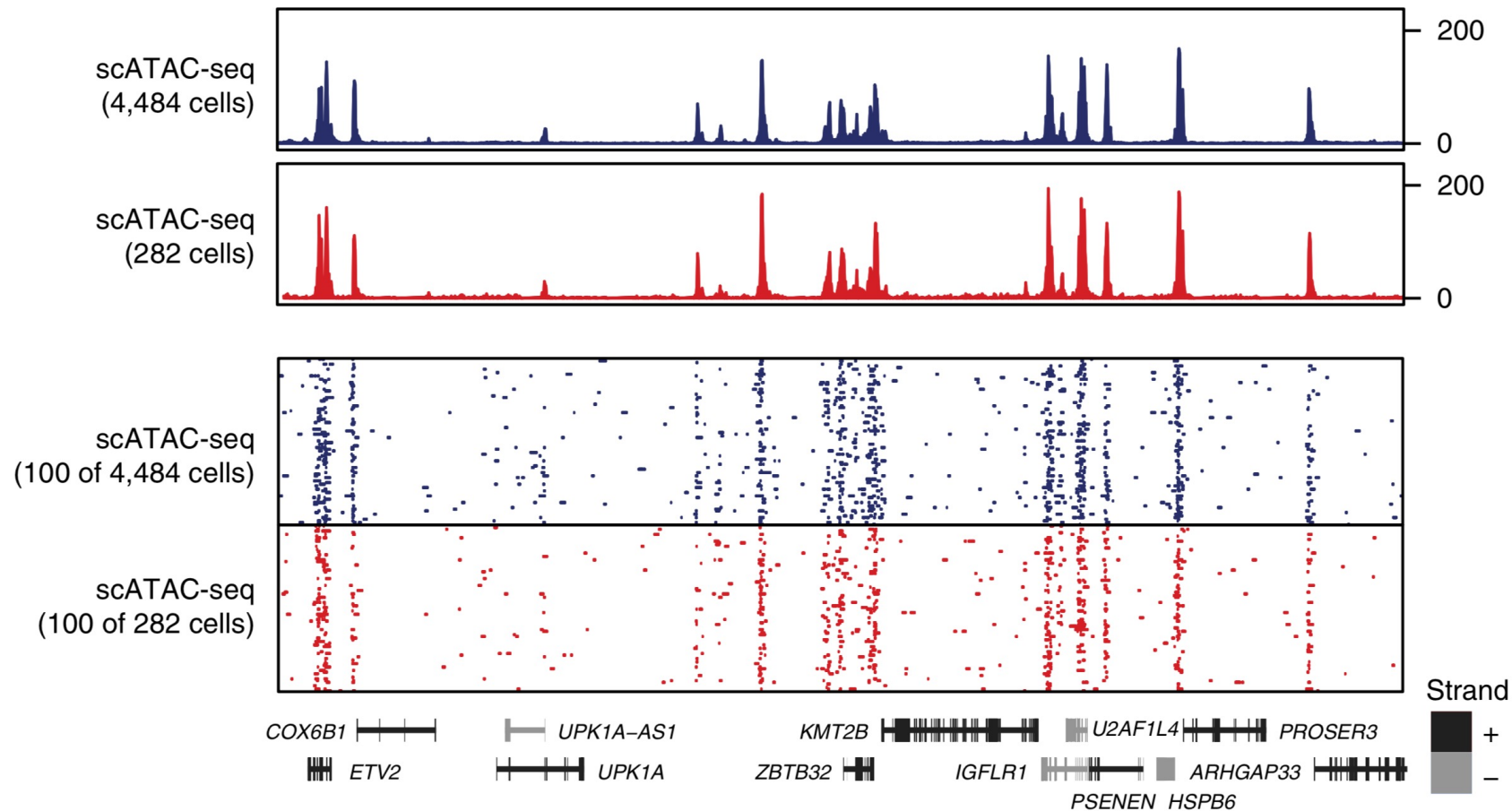
- An alternative to the droplet-based method from 10X genomics is **sci-ATAC-seq** (single-cell combinatorial-indexing with ATAC-seq). (Cusanovich et al. 2015, Science)
 - Here, cells are split up into e.g. 96 wells. Each well has a different short barcode and contains thousands of cells.
 - Cells are then pooled and re-distributed into wells again, adding another short barcode.
 - This is repeated enough times so that each cell will eventually have it's own (almost) unique combination of short barcodes.
- + Low cost per cell, enables high throughput
- Lower cell recovery, important when there is limited starting material



(<https://www.parsebiosciences.com>)

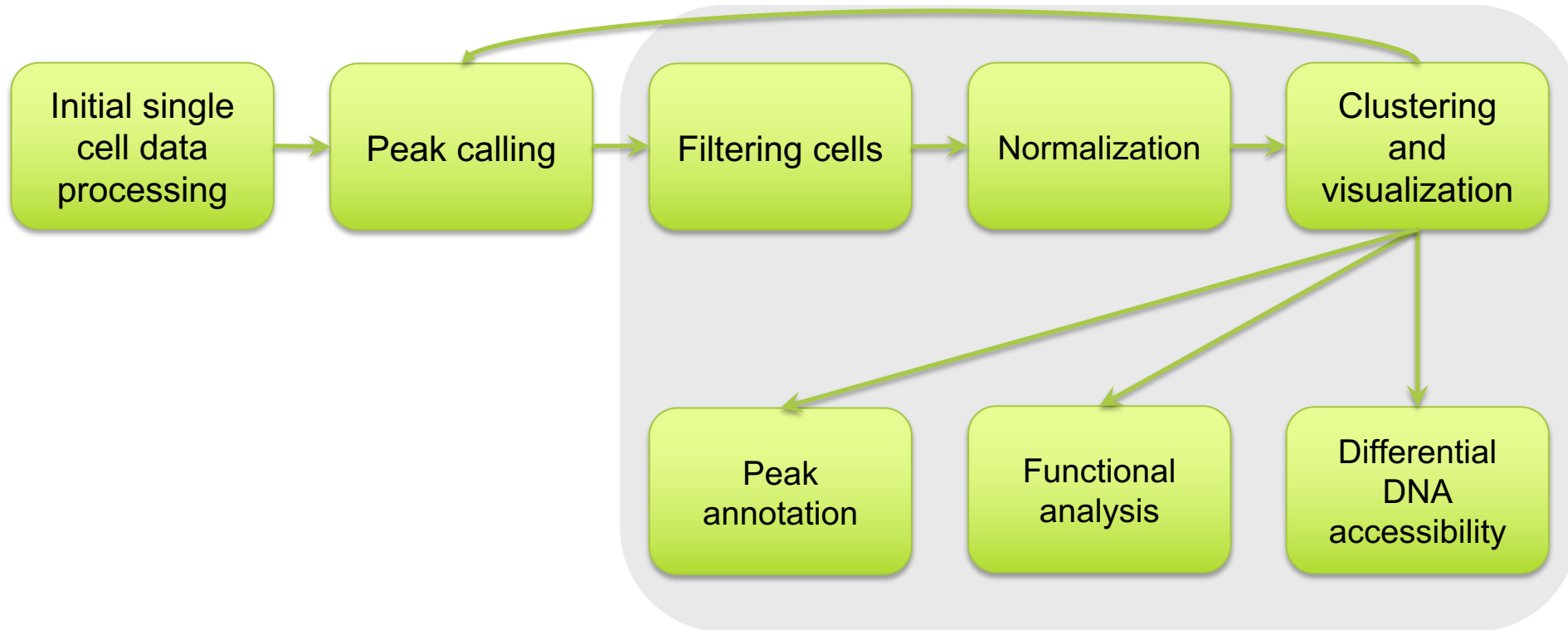
Single cell ATAC-seq data

- Looking at each individual cell, scATAC-seq data are **sparse** and **noisy**.
- But combining data from lots of cells gives meaningful signals.



(Satpathy et al. 2019 Nature Biotechnology)

Single cell ATAC-seq data analysis



Exercise later today

1. Initial single cell data processing

- De-multiplex: Using the cell-specific barcodes, assign each read to a cell.
- (Remove primer sequences.)
- Map reads to the genome, e.g. with **BWA-MEM**.
- Remove duplicates: If several read pairs map to exactly the same coordinates, only one is kept. Such duplicates are assumed to be PCR artifacts.
- Filter out some bad cells already at this stage.

2. Peak calling

- Similar to peak calling for bulk data.
- Done on aggregated data from all cells. (There is not enough data in a single cell to call peaks.)
- If we have a rare cell type with e.g. 50 out of 2000 cells, peaks specific to this cell type can be missed when we use the aggregated data for peak calling.
 - We can go back and redo the peak calling later, only looking at specific groups of cells.
- We then count the reads from every cell in every peak:

	Cell 1	Cell 2	Cell 3	...	Cell M
Peak 1	0	1	1		0
Peak 2	0	0	0		0
Peak 3	0	0	0		1
...					
Peak N	1	0	0		0

← Mostly 0s

3. Filter cells

There are many things that could go wrong in a single cell ATAC-seq experiment

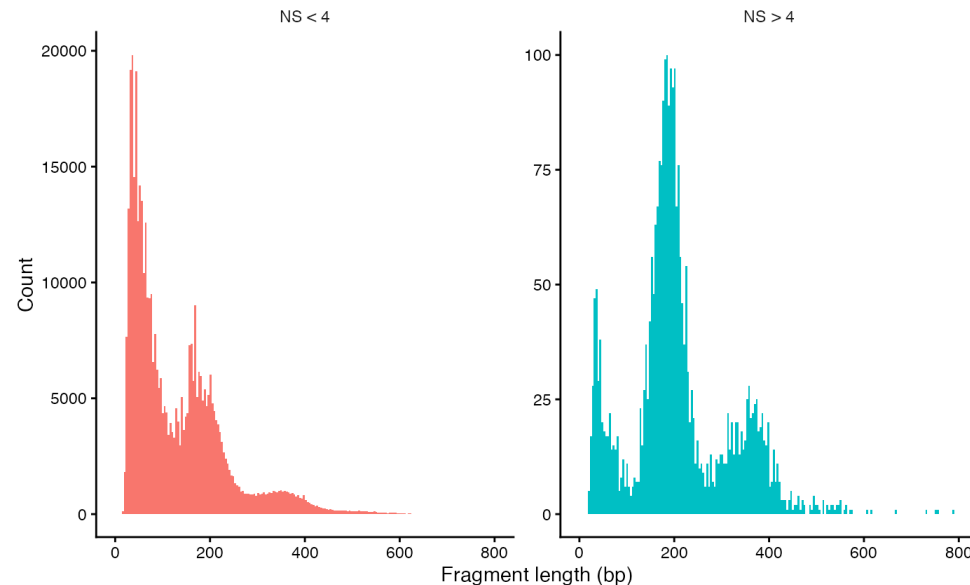
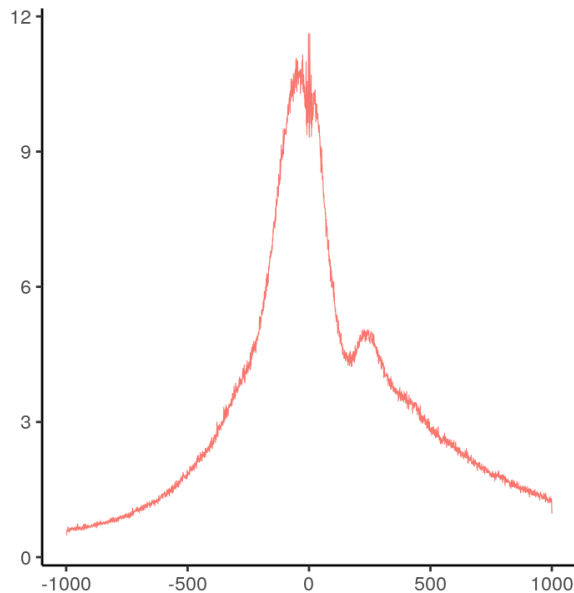
- No cell in a droplet
- Several cells in a droplet
- Dead cells
- Few reads from a cell
- No transfection in a cell

Therefore we use several quality measures to identify and remove problematic cells/barcodes:

- **Number of fragments in peaks.** Cells with very few reads may need to be excluded due to low sequencing depth. Cells with extremely high levels may represent doublets, nuclei clumps, or other artefacts.
- **Fraction of reads in peaks.** Cells with low values (i.e. <15-20%) often represent low-quality cells or technical artifacts that should be removed.

3. Filter cells II

- **Reads in blacklist regions.** The ENCODE project has provided a list of blacklist regions, i.e. regions with artefactual signal. Cells with many reads mapping to these blacklist regions (compared to reads mapping to peaks) often represent technical artifacts and should be removed.
- **Transcriptional start site (TSS) enrichment score.** TSS are associated with open chromatin, so a low level of TSS enrichment would suggest poor ATAC-seq experiments.
- **Nucleosome banding pattern.** The histogram of DNA fragment sizes should exhibit a strong nucleosome banding pattern corresponding to the length of DNA wrapped around a single nucleosome. A high nucleosome signal (ratio of mononucleosomal fragments (147-294 bp) to nucleosome-free fragments (<147 bp)) indicates that a cell has too much wrapped DNA and a low amount of open chromatin (lower data quality, poor cell lysis, or high background noise).



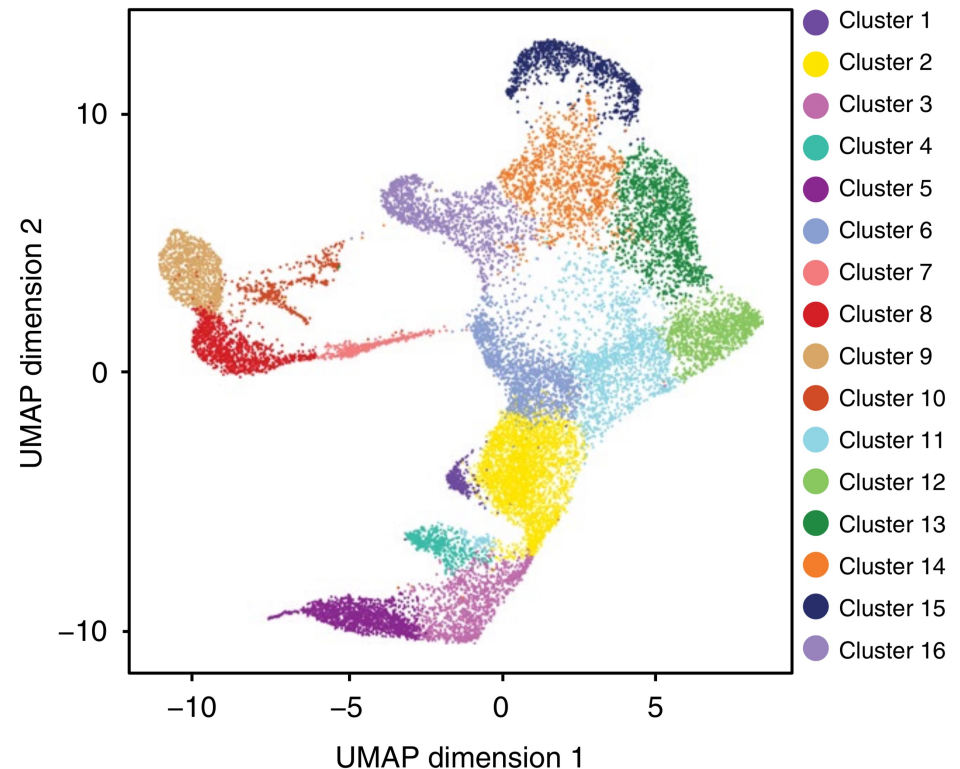
4. Normalization

- Account for different sequencing depth in different cells
- Create a simplified representation of the data, using dimension reduction (singular value decomposition) – similar to principal component analysis.
 - The idea behind this is to reduce noise, and to select informative features to improve clustering of cells and visualization.
 - Typically, the first component correlates with sequencing depth, so by removing it we get rid of artefactual signal.
 - Reducing dimensionality is often good in itself.
 - Results are often better when we select only some features (peaks):
 - Those with highest signal;
 - Those with highest variability.

5. Cluster and visualize cells

- Visualize how cells relate to each other
 - Project many dimensions down to 2.
 - Conceptually similar to PCA, but not linear
 - UMAP algorithm
- Clustering, to identify groups of similar cells (representing different cell types or cell states).

- There are many different clustering methods
- Many settings for such methods
- Trial and error

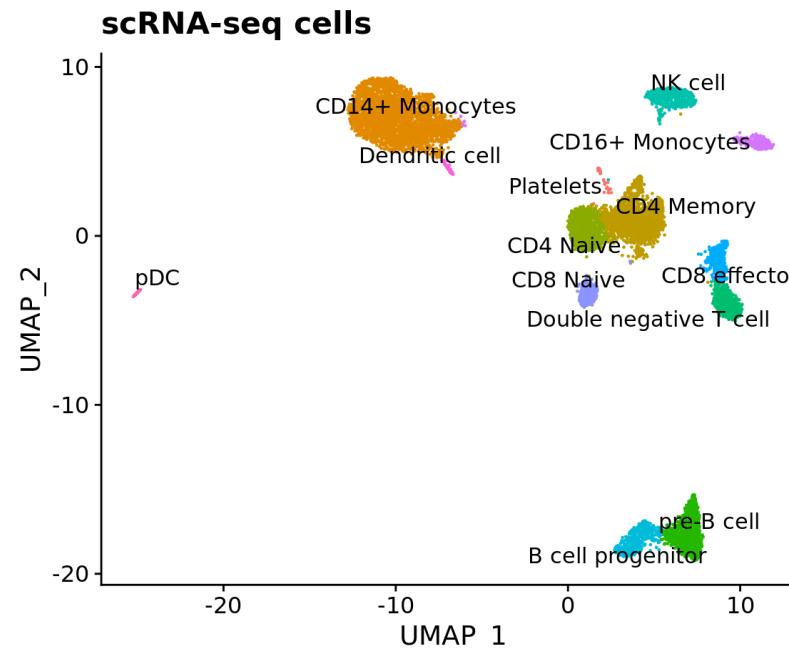
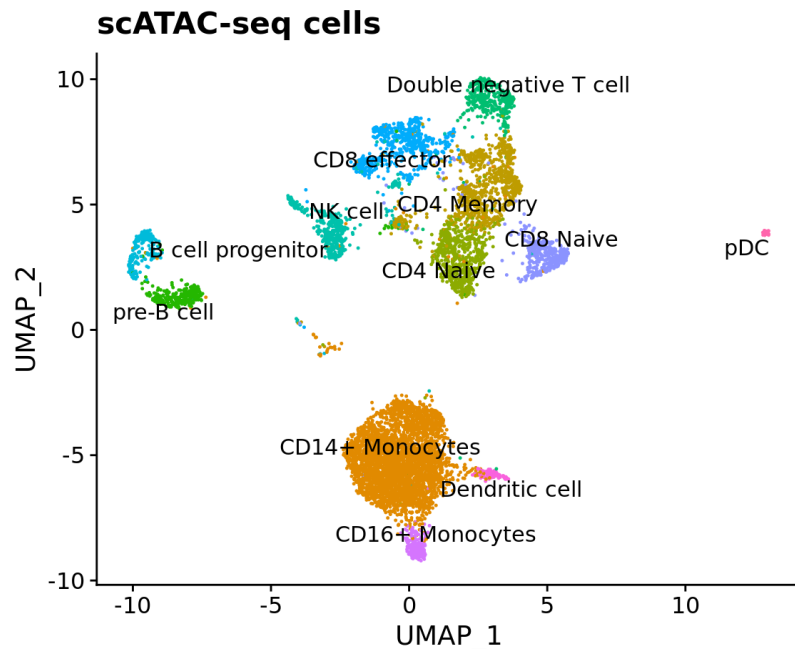


(Satpathy et al. 2019 Nature Biotechnology)

5. Cluster and visualize cells II

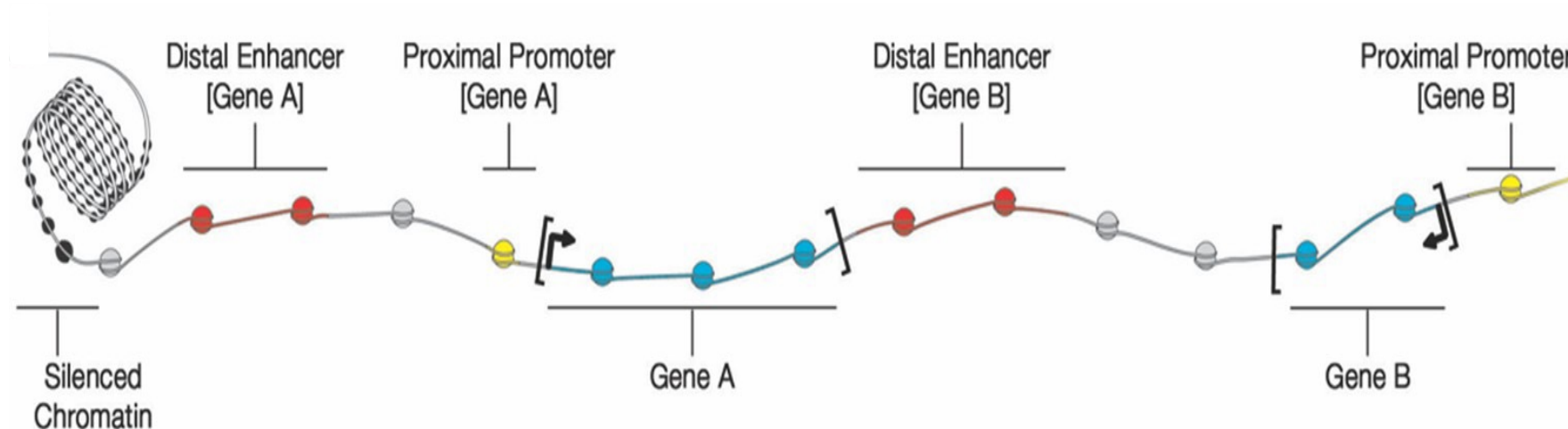
It's often not clear which cell types etc. these clusters represent.

- In single cell RNA-seq we can look at marker genes, unique to a specific cell type. In single cell ATAC-seq, this is harder.
- If it's possible to get RNA-seq data from a similar set of cells, these can be annotated and then used to annotate the ATAC-seq clusters.
- This is called label transfer and will be discussed in the next talk about integrating omics data, and in the exercise.



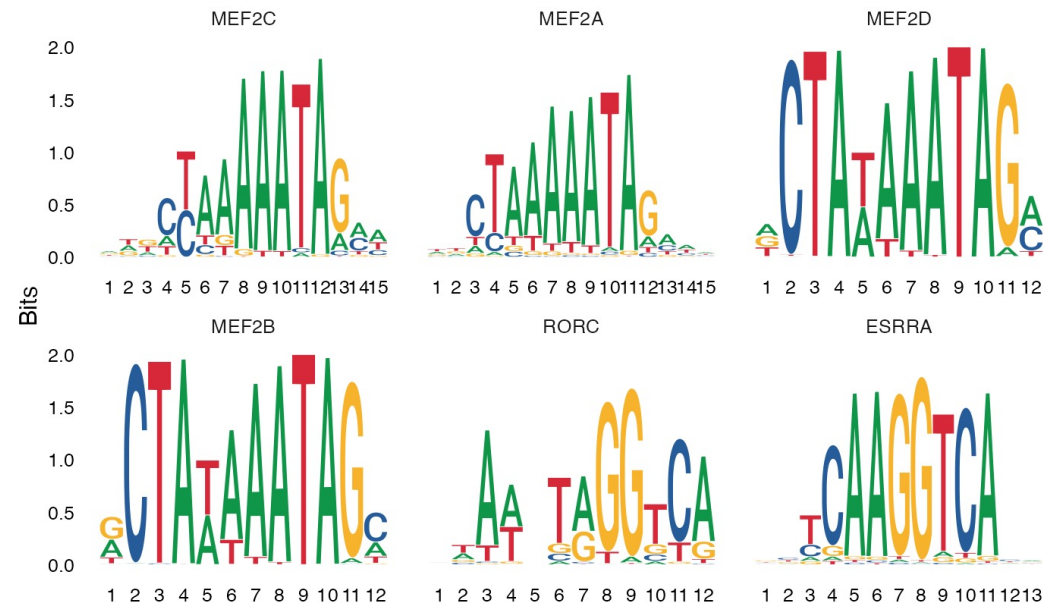
6. Peak annotation

- To easier interpret the peaks, it's useful to note their location with regard to the nearest gene (or the nearest TSS).
 - Remember that a region might not interact with the nearest gene, this is just a starting guess!
- This is similar to what was discussed for ATAC-seq and ChIP-seq data.



7. Functional analysis

- As for bulk ATAC-seq the regions with open chromatin can be further analyzed, to see which transcription factors might bind there. This can give important information on which signaling pathways drive gene expression in different cells.
 - Looking for enriched motifs
 - Cross-referencing open chromatin regions against public ChIP-seq data on different TFs.
- This can be done for each cell or cluster of cells.



8. Differential DNA accessibility

- It is often interesting to know which chromatin regions differ in accessibility between cell types.
- This is a similar problem to differential gene expression (for RNA-seq data) and differential binding (for ChIP-seq data).
- But single cell ATAC-seq data have special properties that make the analysis different from bulk analysis.
 - Sparse data (low read counts, most entries are 0)
 - Many replicates/cells.
- Examples of methods:
 - Logistic regression
 - Negative binomial generalized linear model

Some single cell ATAC-seq analysis tools

ATAC Cell Ranger

Computational pipeline from 10X genomics, does (more or less) all of the analysis steps described here

Seurat/Signac

R packages originally developed for single cell RNA-seq: Filtering cells, normalization, clustering, visualization, differential DNA accessibility and data integration.

SnapATAC (Single Nucleus Analysis Pipeline for ATAC-seq)

SnapATAC2 is a flexible, versatile, and scalable single-cell omics analysis framework that scales to more than 10 million cells.

Scanpy/episcanpy

Python package, originally developed for single cell RNA-seq. Similar functionality to Seurat/Signac

ArchR (Analysis of Regulatory Chromatin in R)

R package for processing and analyzing single-cell ATAC-seq data

(better)ChromVar

R package, mostly useful for motif analysis. (Can do clustering, visualization, differential DNA accessibility too..)

Single cell ChIP-seq-like methods

ChIP on single cells, e.g. using droplets, is hard.

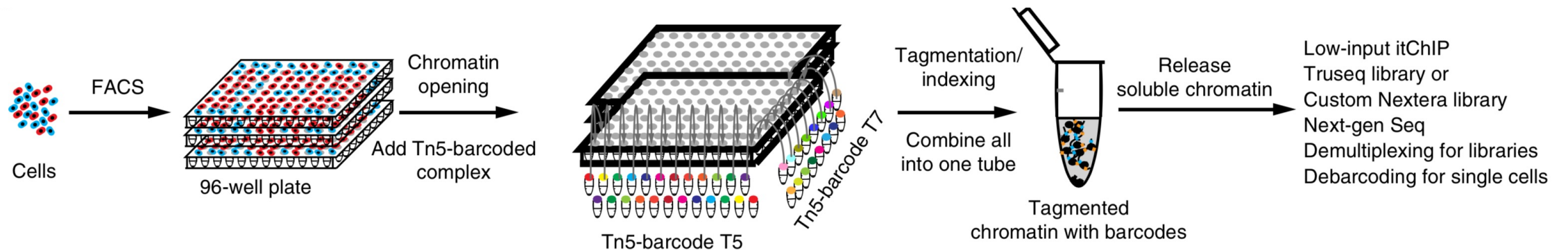
(Rotem et al. 2015, *Nature Biotechnology*) had around 800 reads/cell. Still enough to distinguish different cell types.

(Grosselin et al. 2019, *Nature Genetics*) had around 1600 reads/cell.

Tagmentation based methods:

(Ai et al. 2019 *Nature Cell Biology*) came up with sc-itChIP-seq (single cell indexing and tagmentation ChIP-seq):

- First use Tn5 transposase to add cell barcodes to DNA.
- Then do ChIP in bulk.
- 9000 reads / cell.
- 96 well plates



(Ai et al. 2019 *Nature Cell Biology*)

Single cell ChIP-seq-like methods II

ChIP-free methods:

(Wang et al. 2019, Molecular Cell) CoBATCH

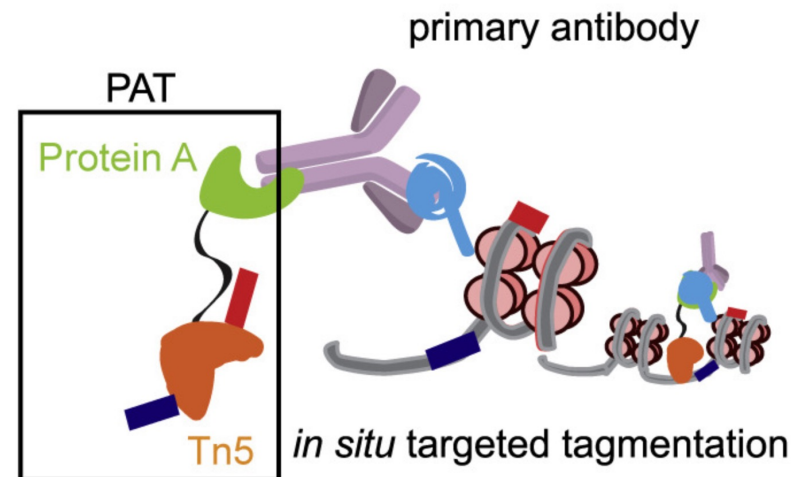
Antibody binds to protein of interest. → This recruits PAT complex with Tn5 → Tagmentation of DNA near protein of interest.
12000 reads/cell

Combinatorial indexing (like for sci-ATAC-seq)

Quite simple protocol, no ChIP

(Kaya-Okur et al. 2019, Nature Communications) CUT&Tag, similar idea.

Used nanowells instead of combinatorial indexing



Single cell ChIP-seq-like methods III

scCUT&Tag: Combined CUT&Tag with droplet-based single-cell library preparation.



Single-cell CUT&Tag profiles histone modifications and transcription factors in complex tissues

Marek Bartosovic¹✉, Mukund Kabbe¹ and Gonçalo Castelo-Branco^{1,2}✉

In contrast to single-cell approaches for measuring gene expression and DNA accessibility, single-cell methods for analyzing histone modifications are limited by low sensitivity and throughput. Here, we combine the CUT&Tag technology, developed to measure bulk histone modifications, with droplet-based single-cell library preparation to produce high-quality single-cell data on chromatin modifications. We apply single-cell CUT&Tag (scCUT&Tag) to tens of thousands of cells of the mouse central nervous system and probe histone modifications characteristic of active promoters, enhancers and gene bodies (H3K4me3, H3K27ac and H3K36me3) and inactive regions (H3K27me3). These scCUT&Tag profiles were sufficient to determine cell identity and deconvolute regulatory principles such as promoter bivalency, spreading of H3K4me3 and promoter-enhancer connectivity. We also used scCUT&Tag to investigate the single-cell chromatin occupancy of transcription factor OLIG2 and the cohesin complex component RAD21. Our results indicate that analysis of histone modifications and transcription factor occupancy at single-cell resolution provides unique insights into epigenomic landscapes in the central nervous system.

Single cell ChIP-seq-like methods IV

- Data analysis for all of these methods is similar to single cell ATAC-seq.
- Single cell DNA-protein interaction studies, i.e. “ChIP-seq-like” are still relatively new, but developing fast and throughput will likely increase a lot.

Protocol | Published: 21 December 2023

Nano-CUT&Tag for multimodal chromatin profiling at single-cell resolution

[José Ramón Bárcenas-Walls](#), [Federico Ansaloni](#), [Bastien Hervé](#), [Emilia Strandback](#), [Tomas Nyman](#),
[Gonçalo Castelo-Branco](#) & [Marek Bartošovič](#) ✉

[Nature Protocols](#) **19**, 791–830 (2024) | [Cite this article](#)

9755 Accesses | **13** Citations | **10** Altmetric | [Metrics](#)

Single cell DNA methylation

Methods

Whole genome vs. reduced representation/targeted
reduced representation/targeted
Bisulphite vs bisulphite-free

Quite hard and expensive

Data

Predominately CpG sites
Thousands of cells
 10^4 - 10^7 CpGs per cell
Not the same CpGs in all cells.

Analysis hard but:

- scDNAm analysis methods are emerging such as scMethCraft.
- Multi-source data integration crucial.

Article

DNA methylation atlas of the mouse brain at single-cell resolution

<https://doi.org/10.1038/s41586-020-03182-8>

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Open access

Hanqing Liu^{1,2,16}, Jingtian Zhou^{1,3,16}, Wei Tian¹, Chongyuan Luo^{1,4}, Anna Bartlett¹, Andrew Aldridge¹, Jacinta Lucero⁵, Julia K. Osteen⁵, Joseph R. Nery¹, Huaming Chen¹, Angeline Rivkin¹, Rosa G. Castanon¹, Ben Clock⁶, Yang Eric Li⁷, Xiaomeng Hou^{8,9,10,11}, Olivier B. Poirion^{8,9,10,11}, Sebastian Preissl^{8,9,10,11}, Antonio Pinto-Duarte⁵, Carolyn O'Connor¹², Lara Boggeman¹², Conor Fitzpatrick¹², Michael Nunn¹, Eran A. Mukamel¹³, Zhuzhu Zhang¹, Edward M. Callaway¹⁴, Bing Ren^{7,8,9,10,11}, Jesse R. Dixon⁵, M. Margarita Behrens⁵ & Joseph R. Ecker^{1,15,52}

Article | [Open access](#) | Published: 15 May 2026

Dissecting epigenetic heterogeneity in single-cell DNA methylomes with a unified framework

[Songming Tang](#), [Siyu Li](#), [Guangxin Zhang](#), [Aoran Lyu](#), [Han Li](#) & [Shengquan Chen](#) 

[Nature Communications](#) **17**, Article number: 6469 (2026) | [Cite this article](#)

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Combining assays from the same cells

Many methods combine several assays from the same cells

scRNA-seq and scATAC-seq (10X genomics, SNARE-seq, and many more)

scRNA-seq and sc-protein abundance (CITE-seq)

scRNA-seq and scDNA methylation

scRNA-seq and scDNA methylation and sc nucleosome (scNMT-seq)

scRNA-seq, scATAC-seq, sc-protein abundance and clonal info from mitochondrial DNA (DOGMA-seq)

Method	Molecular layers profiled				Throughput (low/medium/ high)	Special features (compared to techniques from same category)	Format	References	
	Epigenome		Genome	Transcriptome					
	Chromatin accessibility	Chromatin conformation							DNAme
scCAT-seq	x				x	+	↑ usable fragments	well	Liu et al. (2019)
Paired-seq	x				x	+++	↑ throughput	well	Zhu et al. (2019)
sc(ATAC + RNA)-seq	x				x	+	↓ cost; simple workflow	well	Reyes et al. (2019a)
sci-CAR	x				x	+++	↑ acc. & RNA intersect coverage	well	Cao et al. (2018)
SNARE-seq	x				x	+++	↑ sensitivity	droplet	Chen et al. (2019)
ASTAR-seq	x				x	++	↓ price-performance ratio	microfluidics	Xing et al. (2020)
SHARE-seq	x				x	+++	↑ throughput, performance	well	Ma Sai. et al. (2020)
ISSAAC-seq	x				x	+++	↑ throughput, performance (esp. ATAC)	well/droplet	Xu et al. (2022)
scDam&T-seq		x			x	+	protein-DNA interactions information	well	Rooijers et al. (2019)
scNOMe-seq	x		x			+	estimates nucleosome phasing	well	Pott, (2017)
scCOOL-seq	x		x	x		+	↑ acc. & DNAme intersect coverage	well	Guo et al. (2017)
iscCOOL-seq	x		x			++	↑ accessibility coverage	well	Gu et al. (2019a)
scMethyl-HiC		x	x			+	↑ mapping rate	well	Li et al. (2019)
sn-m3C-seq		x	x			+++	↑ DNAme coverage	well	Lee et al. (2019)
scNMT-seq	x		x		x	++	↑ throughput	well	Clark et al. (2018)
scNOMeRe-seq	x		x		x	+	↑ DNAme coverage	well	Wang et al. (2021)
scSIDR-seq				x	x	+	captures total RNA	well	Han et al. (2018)
TARGET-seq				x	x	+++	↓ cost; ↑ throughput	well	Rodriguez-Meira et al. (2019)
RETrace			x			+	captures microsatellites	well	Wei and Zhang, (2020)
scTrio-seq2			x	x	x	++	↑ DNAme coverage	well	Bian et al. (2018)



Spatial methods

Sequencing based methods – “spatial transcriptomics”

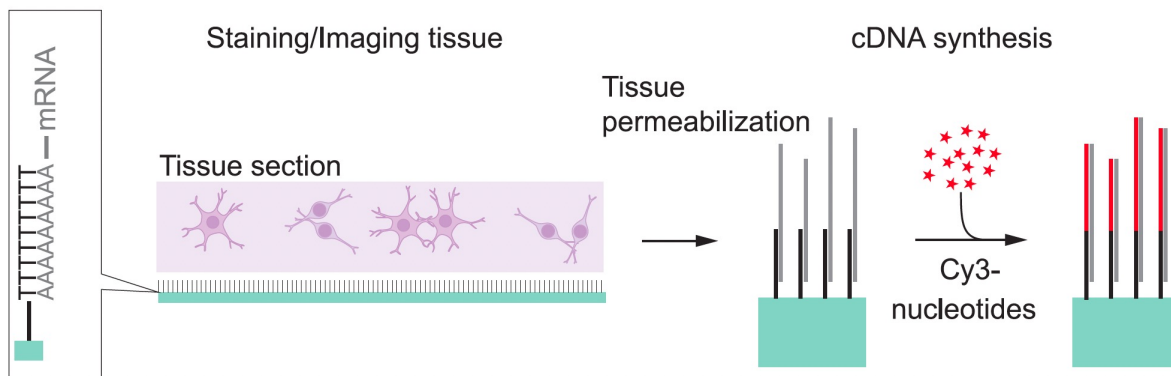
- + captures all genes
- each spot does not correspond to a single cell.

The basic idea is to hybridize a tissue to an array, where spatial barcodes are added to the RNA molecules-

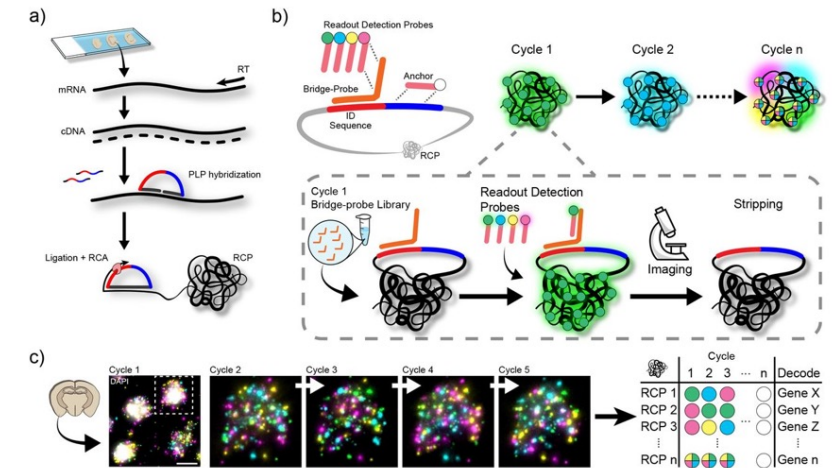
Imaging based methods – “in-situ sequencing”

- probes a subset of genes
- + subcellular resolution

Targeted approach, using padlock probing



Ståhl et al 2016, Nature



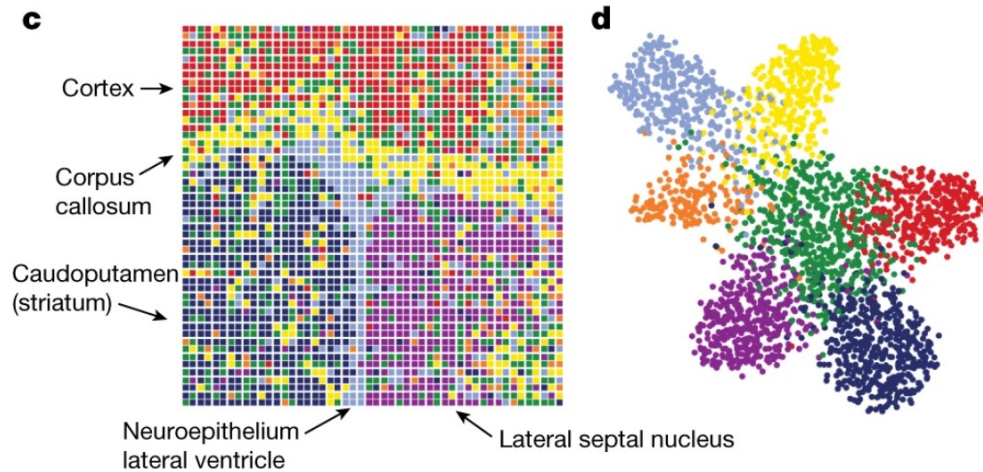
Gyllborg et al, Nucleic Acids Research (2020)

Spatial methods, II

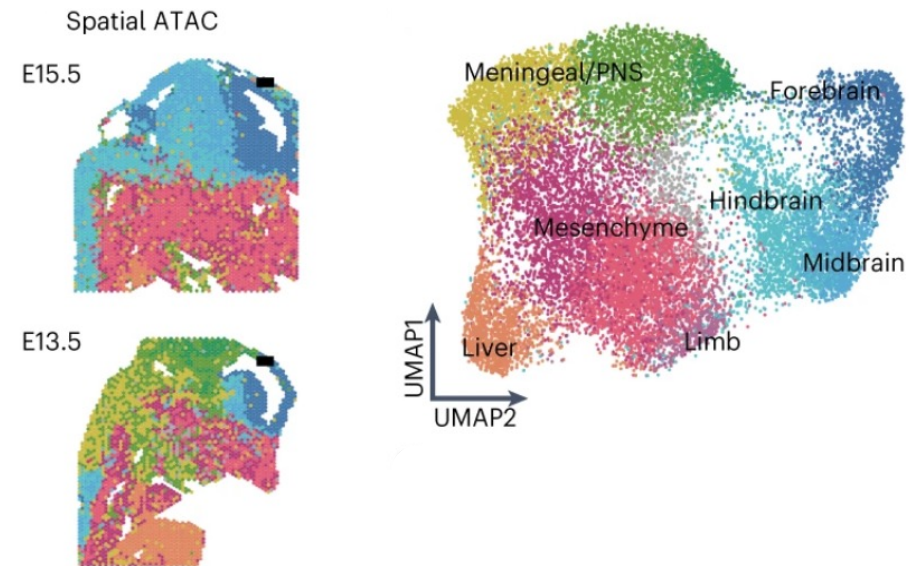
- For spatial transcriptomics, a lot of the analysis is similar to for single cell data, but instead of cells, you work with “spots” (and these may cover several cells).
- Spatial methods are often used together with single cell methods, to get information both on what different cell types looks like (from the single cell data) and where they are located (from the spatial data)

Spatial methods, III

Spatial methods have been adapted to ATAC-seq (in situ Tn5 transposition).



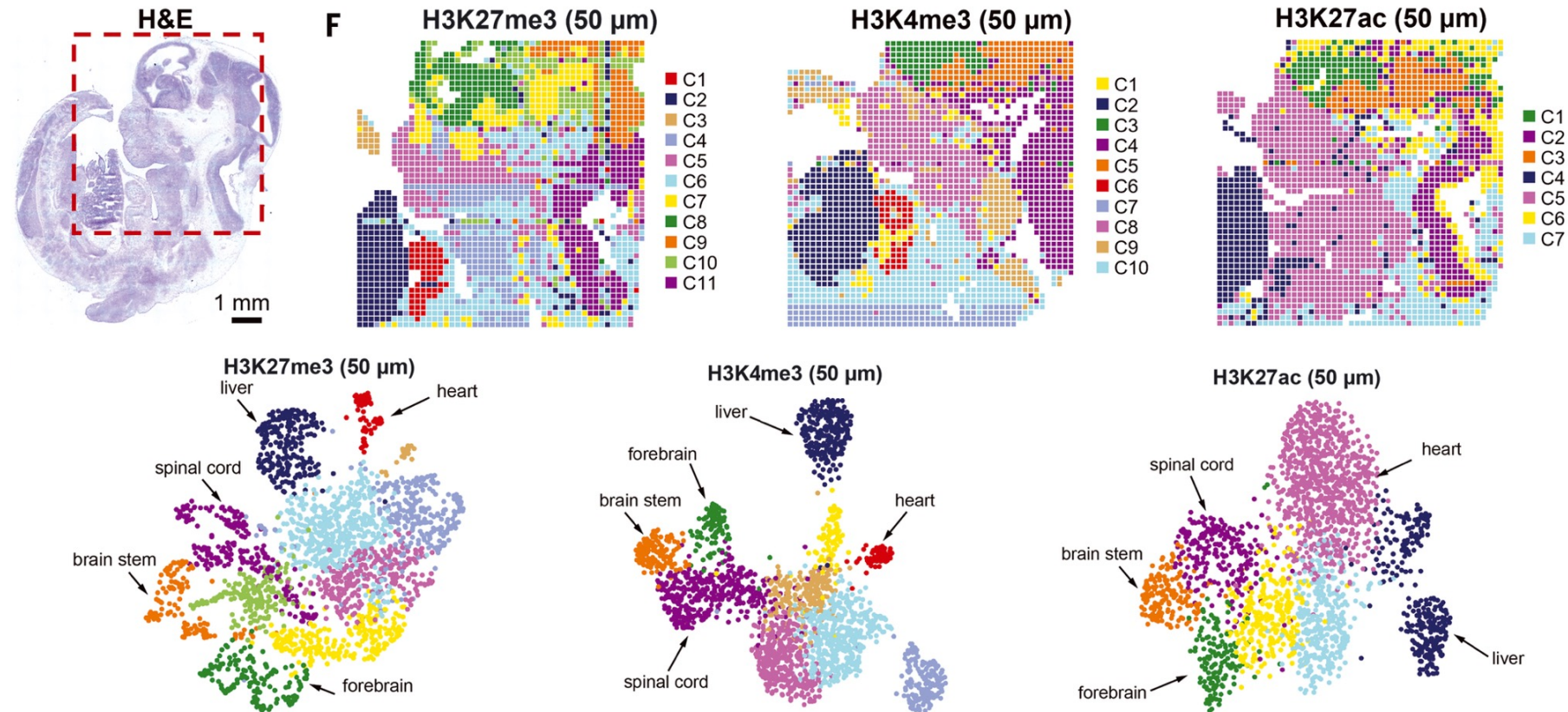
Deng et al 2022, Nature



Llorens-Bobadilla et al 2023, Nature Biotech.

Spatial methods, IV

Spatial methods have also been adapted to CUT&Tag (in situ CUT&Tag chemistry).



Deng et al 2022, Science

Summary

Single cell ATAC-seq

Usually works well

Commercial kits available

You will have a look at the data analysis in the exercise.

Single cell ChIP-like

Single-cell CUT&Tag works quite well.

Mostly for histone marks, seems more challenging for TFs.

Single cell DNA methylation

A lot of development happening

Useful methods will become more widely available (already scWGBS/splat-seq at NGI).

Single cell spatial methods

Well established for transcriptomics

Adaptations for epigenomics are being developed, but still not widely used.

Integration of different types of genomics data

Markus Ringnér

SciLifeLab, NBIS, Lund University

Workshop – Epigenomics Data Analysis

Data integration

Combine data from different studies and different technologies

RNA-seq

ATAC-seq

DNA-methylation

Spatial transcriptomics

Proteomics

Metabolomics

..

The goal: Learn something you couldn't learn from looking at each individual data set independently

Some data integration applications

See how different data sets correlate

- TF binding vs gene expression
- Two TF binding data sets
- Gene expression vs open chromatin

Learn something about one data set from looking at another

- Cell type composition
- Label transfer
- Deconvolution.

Use several data sets to predict phenotype

- supervised learning

...

This talk will be a pretty general discussion on what data integration can be used for and some methods to do this.

How do we want to combine data?

Combine data sets with the **same features** in **different samples**.

	Sample 1	Sample 2	...	Sample M		Sample M+1	Sample M+2	...	Sample M+K
Gene 1					Gene 1				
Gene 2					Gene 2				
..					..				
Gene N					Gene N				

Remove batch effects

Combine data sets with **different features** in the **same samples**.

	Sample 1	Sample 2	...	Sample M
Gene 1				
Gene 2				
..				
Gene N				

	Sample 1	Sample 2	...	Sample M
CpG 1				
CpG 2				
..				
CpG L				

← Makes some analysis harder

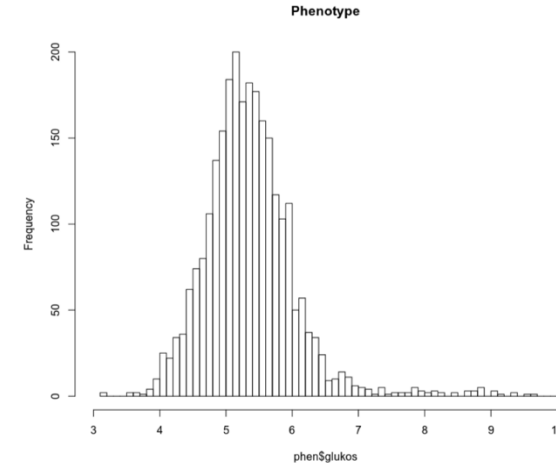
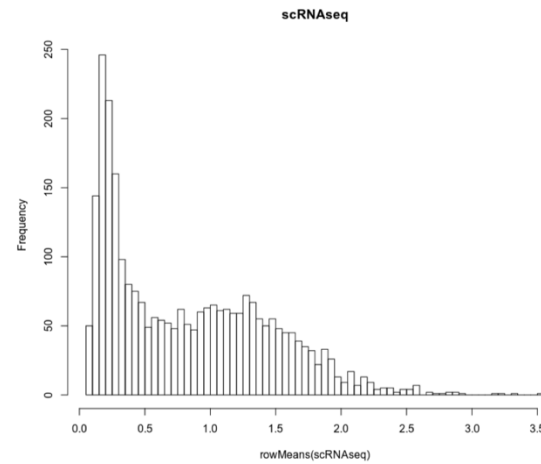
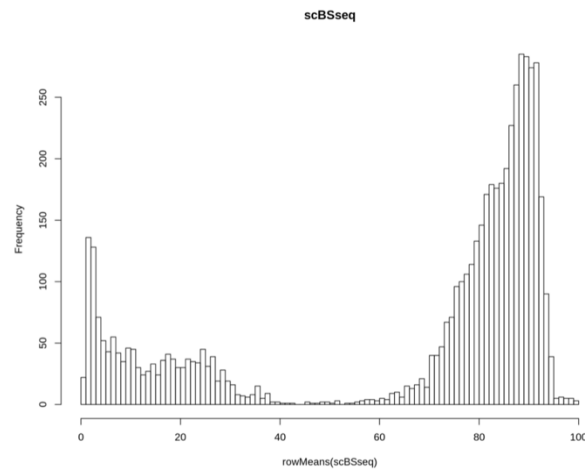
Combine features

Not trivial how to combine data from different assays

Different number of features

Different biases

Different distributions



Integration of genomic ranges

Many types of data can be represented as regions across the genome

Genes

Promoters

CpG islands

Open chromatin

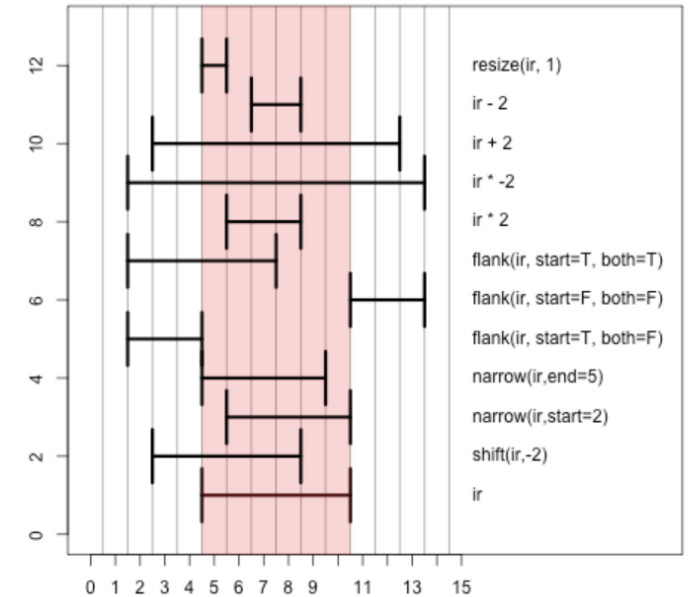
Transcription factor binding sites

SNPs

R package to manipulate genomic ranges, **GRanges**.

Underlies a lot of tools in R.

Bedtools has similar functionality



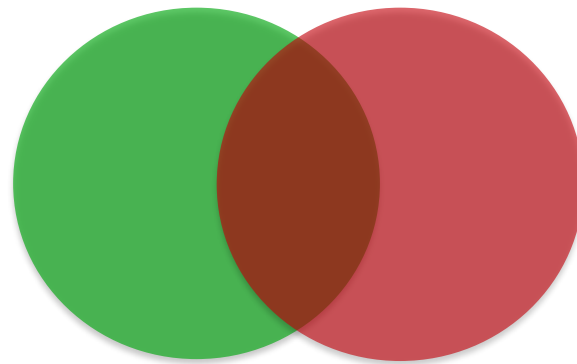
Integration of genomic ranges II

A common application is to test if the overlap between two data sets is statistically significant (e.g. greater than expected by chance).

If the binding sites of two transcription factors overlap

If a protein preferably binds around transposable elements, promoter regions, CpG islands etc.

...



Integration of genomic ranges III

Many ways to do this, many ways to model what is expected by chance.

Theoretical (fast) vs empirical (many randomizations, slow)

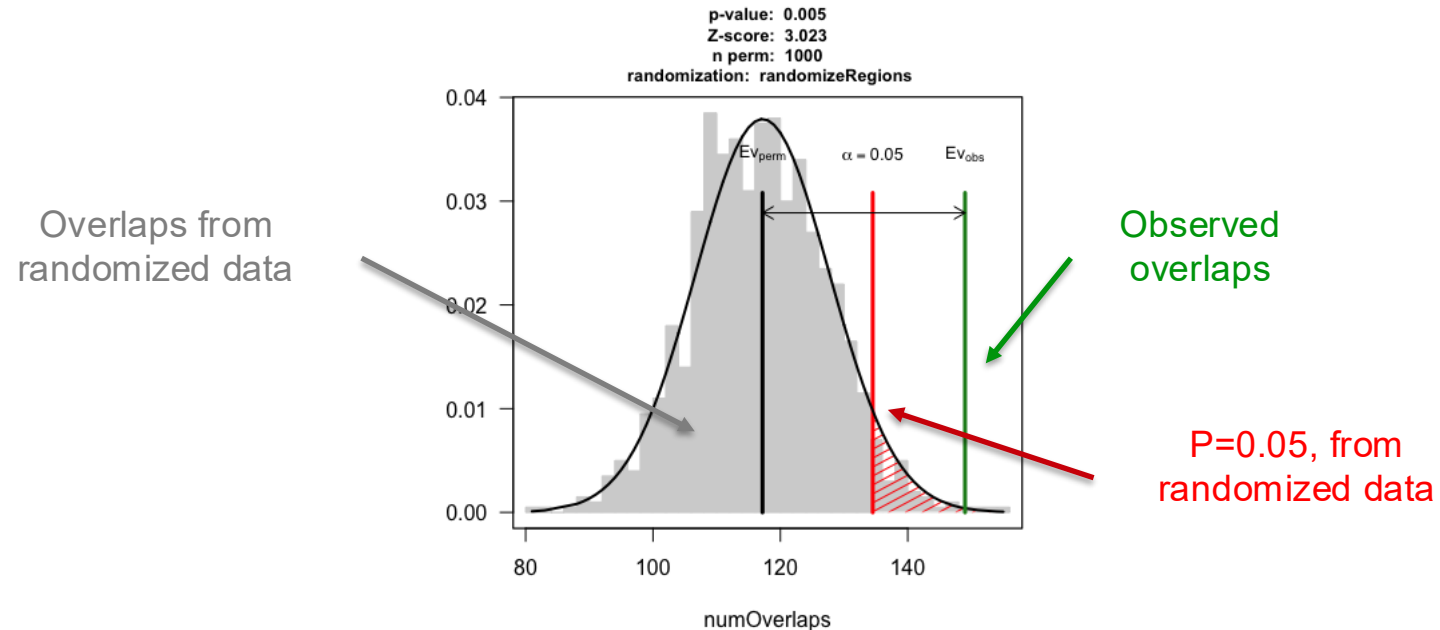
Take region size into account?

Take distances between regions into account?

Take linkage disequilibrium into account?

Do the lengths of the overlaps influence the result?

This can have a big impact on results.

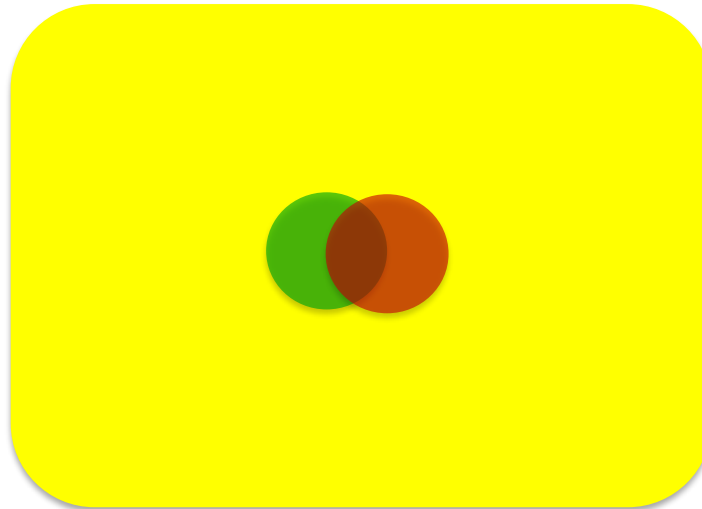


Integration of genomic ranges IV

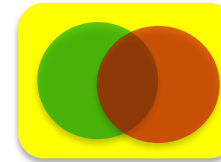
The most important difference between methods is the null distribution

How much overlap do we expect by chance?

This depends a lot on the choice of universe, e.g. the whole genome or just promoters.



Big universe (e.g. whole genome)
→ significant overlap



Small universe (e.g. promoters)
→ same overlap is not significant

Integration of single cell data

It is often useful to integrate different single cell data sets. Could be

- from different batches

- from different studies

- your data with public atlas

- RNA-seq with ATAC-seq

Some have the same features, but combine cells. Others have the same cells, but combine features.

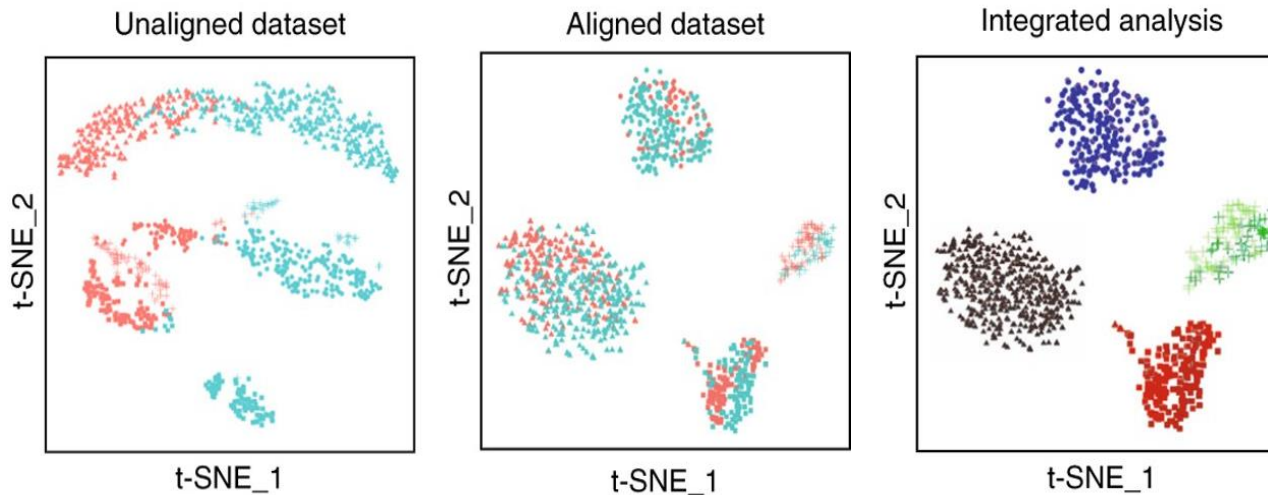
Some version of this is used in many/most single cell studies.

Integration of single cell data, II

Integrate data so that cells from different experiments (and omics) can be analyzed together.

Make sure measurements are on the same format

Align data sets

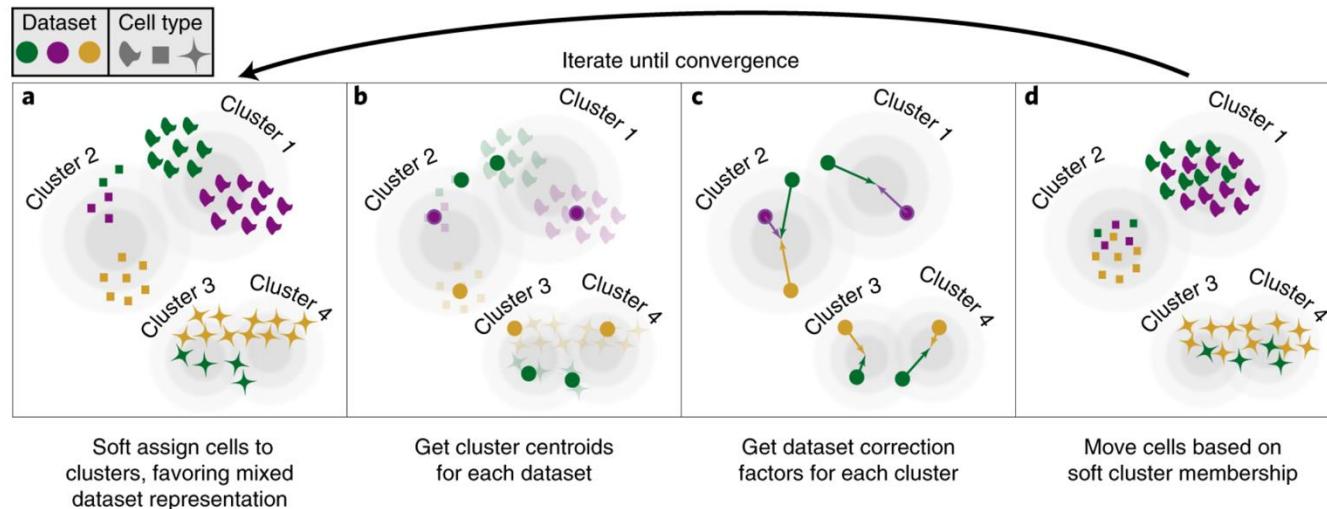


(Butler et al. 2018, Nature Biotechnology)

Data integration with Harmony

Assumption: Clusters, representing cell types, should contain cells from many/all batches. →
Compute batch correction model based on clustering.

Korsunsky et al. 2019, Nature Biotech



PCA embeds cells into a space with reduced dimensionality. Harmony accepts the cell coordinates in this reduced space and runs an iterative algorithm to adjust for dataset specific effects. **a**, Harmony uses fuzzy clustering to assign each cell to multiple clusters, while a penalty term ensures that the diversity of datasets within each cluster is maximized. **b**, Harmony calculates a global centroid for each cluster, as well as dataset-specific centroids for each cluster. **c**, Within each cluster, Harmony calculates a correction factor for each dataset based on the centroids. **d**, Finally, Harmony corrects each cell with a cell-specific factor: a linear combination of dataset correction factors weighted by the cell's soft cluster assignments made in step **a**. Harmony repeats steps **a** to **d** until convergence. The dependence between cluster assignment and dataset diminishes with each round. Datasets are represented with colors, cell types with different shapes.

Seurat 5 supports a multitude of data integration methods including Harmony

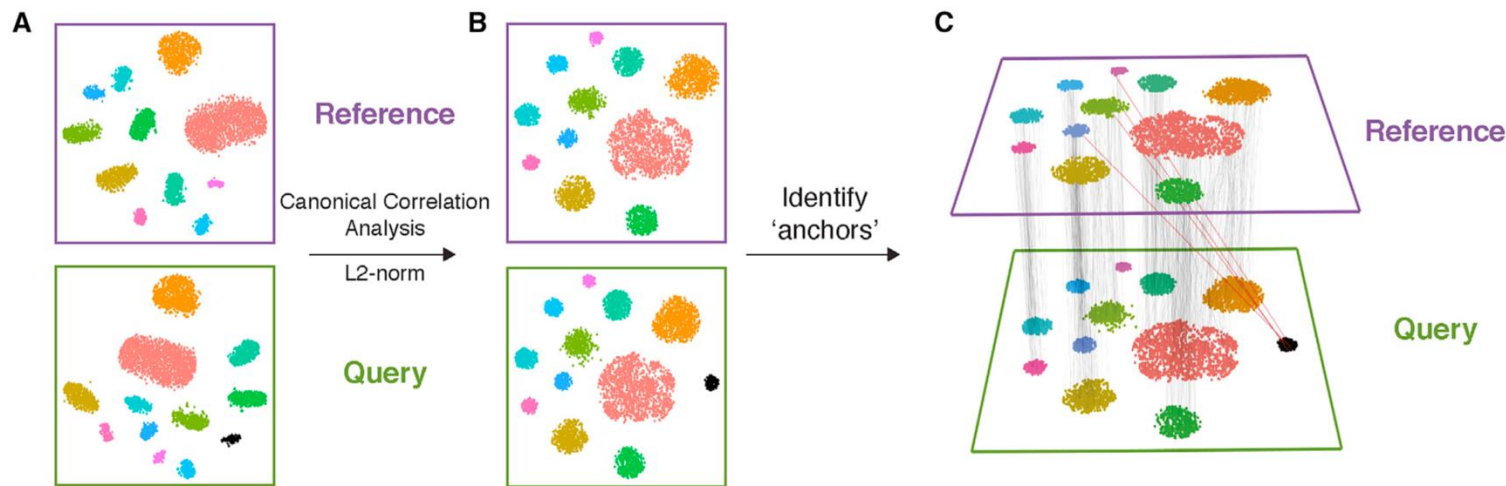


Label transfer from scRNA-seq to scATAC-seq

Label transfer: Use cell type annotations from one data set (e.g. RNA) to annotate another data set (e.g. ATAC).

Brief outline of process in Signac (Seurat)

1. Calculate gene activity on scATAC-seq data set
2. Normalize both data sets
3. Use CCA method to project both data sets to a common space
4. Find “anchors”, e.g. pairs of cells from the scRNA-seq and scATAC-seq data sets that are mutual nearest neighbors (MNNs)
5. Transfer labels to all scATAC-seq cells by inference from its nearest anchor cell.



You will use Signac/Seurat to integrate scATAC-seq with scRNA-seq data, and do label transfer in an exercise today.

Supervised methods for data integration

Build models that can combine several data sets to predict an outcome or another dataset.

Example 1: Use RNA-seq + DNA methylation + H3K27me3 to predict disease outcome

Is prediction using all data sets together better than prediction with each individual data set?

Example 2: Use ATAC-seq to predict RNA-seq

The model itself can be interesting to investigate: which features are important, which features interact.

Curse of dimensionality

Often many features and few samples

Feature selection crucial

Many different types of methods

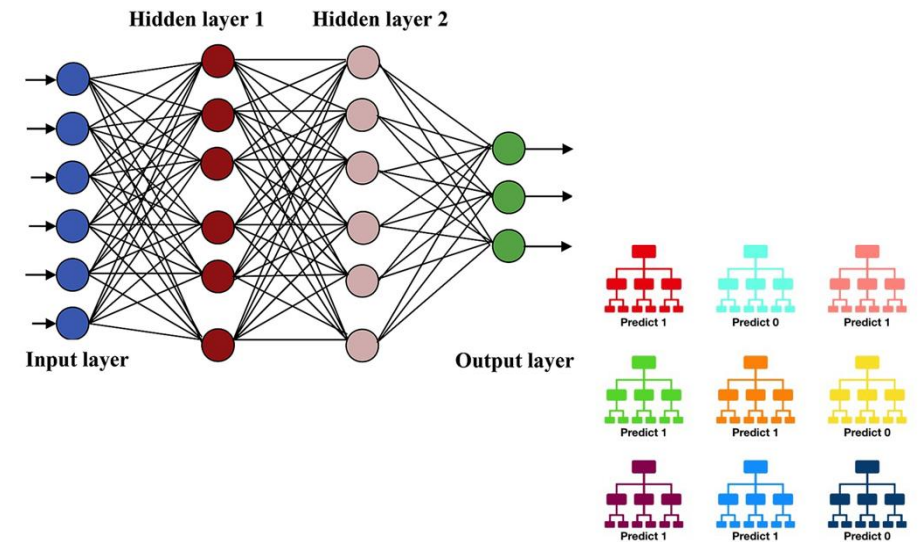
Random Forest

LASSO/Ridge regression etc.

PLS/PLS-DA/OPLS

Neural networks

This is a very big topic, outside the scope of this course.



Summary

Many different goals of data integration and many ways to do it

Simple comparisons between data sets (how well do the data sets agree/correlate/overlap?)

Integration of single cell data (label transfer etc.)

Supervised prediction

Example goal: which genes are expressed (RNA-seq), which TFs might bind them (ATAC-seq), and where are these cells located (spatial transcriptomics).

High-level vs. low-level integration

This is a big subject

We have a course on more of this “Omics Integration and Systems Biology”